



2026 BASIC ABSTRACT ALGEBRA STUDY

LECTURE 1 HOMEWORKS

SOLUTION - Q11

Version

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Read this carefully before moving on.

Please, clearly specify whether the statement in question is *true* or *false*, for the *prove* or *disprove* type of questions. As always, it is highly encouraged for you to write each step clearly with full sentences.

As a demonstration of a proof answer, consider the following example.

The above statement is **true**.

Proof. ...



As a demonstration of a disproof answer, consider the following example.

The above statement is **false**.

Proof. ...



Note 1 (Notations)

The following notations will be used throughout the following questions.

- $\mathbb{Z}$: A set of all integers.
- $\mathbb{Z}^+$: A set of all positive integers. i.e., $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.
- $\mathbb{Z}_n$: A set of non-negative integers from 0 to $n - 1$ for some positive integer n . For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $n\mathbb{Z}$: A set given as $\{nx \mid x \in \mathbb{Z}\}$ for an integer n . i.e., the set of all multiples of n in $\mathbb{Z}$. For instance, $5\mathbb{Z} = \{\dots, -10, -5, 0, 5, 10, \dots\}$.
- $\mathbb{Q}$: A set of all rational numbers.
- $\mathbb{R}$: A set of all real numbers.
- $\mathbb{R}^+$: A set of all positive real numbers.
- $\mathbb{C}$: A set of all complex numbers.
- $A + B$: A set given as $\{a + b \mid a \in A, b \in B\}$. For instance, $\mathbb{Z}_2 + \mathbb{Z}_3 = \{(0 + 0), (0 + 1), (0 + 2), (1 + 0), (1 + 1), (1 + 2)\} = \{0, 1, 2, 3\}$

Definition 2 (Subset)

Given two sets A and B , A is said to be a subset of B if the following condition holds.

If x is an element of A , then x is an element of B .

Definition 3 (Extensionality)

Two sets A and B are said to be equal if and only if A is a subset of B and B is a subset of A . This is called the *axiom of extensionality*. If two sets A and B are equal, we write $A = B$.

Q1.

EASY

Let A and B be two sets. Let $A(x)$ mean x is an element of A , and $B(x)$ mean x is an element of B .

Which of the following is equivalent to saying that $A \subset B$?

- ☐ 1. $B(x) \rightarrow A(x)$
☐ 2. $\text{not } A(x) \rightarrow B(x)$
☐ 3. $\text{not } A(x) \rightarrow \text{not } B(x)$
- ☐ 4. $A(x) \rightarrow \text{not } B(x)$
☐ 5. $\text{not } B(x) \rightarrow \text{not } A(x)$

Definition 4 (Roots of unity)

Let μ_n denote the set of all complex solutions to an equation of form $x^n = 1$ for a positive integer n . Elements of such a set are called n -th roots of unity.

For instance, $\mu_2 = \{1, -1\}$.

Q2.

EASY

Show that $\mu_2 \subset \mu_4$.

Q3.

EASY

Prove or disprove that $\mathbb{Z} \subset n\mathbb{Z} + m\mathbb{Z}$ for any two distinct positive integers n and m .

Hint 5. Provide concrete n and m that make up a counterexample. Try justifying your answer by using the definition of set builder notation.

Q4.

HARD

Prove or disprove that $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$.

Hint 6. cf. lecture note.

Q5.

HARD

Show using Definition 2 that for any two positive integers n and m , $\mu_n \subset \mu_{nm}$.

Hint 7. Of course, it is a generalization of Q2.

Q6.

HARD

Show by Definition 3 that $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$ for any positive integer n .

Hint 8. cf. PCSA Q4.

HARD

Q7.

Prove or disprove that if $\mathbb{Z} = n\mathbb{Z}$ for an integer n , then either $n = 1$ or $n = -1$.

Hint 9. What are the factors of 1 and -1 ?

Definition 10 (Symmetric difference)

$A \triangle B$ is a binary set operation given by $(A \setminus B) \cup (B \setminus A)$ and is called a *symmetric difference* between A and B .

Q8.

LEVI

Problem by @escentia07.

Prove or disprove that $A \triangle B \subset C$ if and only if $A \cup B \subset (A \cap B) \cup C$.

Q9.

LEVI

For an integer n , let $M(k)$ denote the set $\{x \mid x \equiv k \pmod{n}\}$. Let P be the set given by $P = \{M(k) \mid k \in \mathbb{Z}\}$.

(a) Show that every two distinct elements of P are disjoint.

Hint 11. Use Definition 3 and *contraposition*.

(b) Show that the following holds for any two integers a and b by Definition 3.

$$M(a) + M(b) = M(a + b)$$

Q10.

LEVI

Problem by @escentia07.

Let $f(x) = x^{2121} + x - 1$

How many real solutions does the equation $f^{2026}(x) = x$ have? Justify your answer.

Answer: _____

Q11.

LEVI

Let T be the set $\{z \in \mathbb{C} \mid |z| = 1\}$ and μ_∞ be the set of all roots of unity as follows.

$$\mu_\infty = \bigcup_{n=1}^{\infty} \mu_n = \mu_1 \cup \mu_2 \cup \dots$$

Prove or disprove that $T = \mu_\infty$.

The statement is **× false**.



If you have learned set theory, you might have immediately realized that T is uncountable while μ_∞ is countable.

Here goes the proof!

Proof. Let's first ensure that μ_∞ is countable. We know that each μ_n is finite as $|\mu_n| = n$ and not empty as they always include 1, making μ_∞ not empty. Now take any function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mu_\infty$ such that if $m \leq n$, then $f(n, m)$ is the m -th element of μ_n when ordered by arguments. As such f always exists and is clearly surjective, we have $|\mu_\infty| \leq |\mathbb{Z}^+ \times \mathbb{Z}^+| = \aleph_0$. Hence, μ_∞ is countable. On the other hand, T is uncountable as the argument $\arg : T \rightarrow [0, 2\pi)$ is bijective, and $[0, 2\pi)$ is uncountable. Since one is countable while the other is uncountable, they cannot be equal. $\square$

Note that the above proof is perfectly valid even though it doesn't provide any counterexample. Nevertheless, I provide the below 'counterexample-based' disproof as well.

Proof. Let $x = \frac{4}{5} + \frac{3}{5}i$. It is clear that $|x| = 1$ hence $x \in T$. You may have already realized that this counterexample is from Pythagorean numbers.

If x is an element of μ_∞ , there must exist a positive integer n such that $x^n = 1$, otherwise put, $(4 + 3i)^n = 5^n$. Now, let $(4 + 3i)^k = A_k + B_k i$ for some $k \in \mathbb{Z}^+$ and find the next power.

$$(4 + 3i)^{k+1} = (A_k + B_k i)(4 + 3i) = (4A_k - 3B_k) + (3A_k + 4B_k)i$$

Thus, $A_{k+1} = 4A_k - 3B_k$ and $B_{k+1} = 3A_k + 4B_k$. As $A_1 = 4$ and $B_1 = 3$, we can ensure that every A_k, B_k are integers by induction. From here, it is not hard to show that $A_k \equiv 3B_k \pmod{5}$ and $B_{k+1} \equiv 3B_k \pmod{5}$. As $B_1 = 3$, we can conclude that every B_k is essentially congruent to a power of 3, namely $B_k \equiv 3^k \pmod{5}$.

Since 5^n is a real number, its imaginary part, B_n , must be 0 when $k = n$. However, the congruence equation $3^n \equiv 0 \pmod{5}$ does not have a solution, as $3 \not\equiv 0 \pmod{5}$. Thus, such positive integer n does not exist, contradicting with the assumption. Therefore, x cannot be an n -th root of unity for any positive integer n . $\square$

Keep in mind that multiplication of a complex number is a *rotation* on the complex plane. Thus, what we've just shown here is that the angle (*argument*) of x does not divide the circle evenly. Observe, or at least, feel how the problem of evenly dividing a circle turns into a problem of solving a solution to a

modular congruence. The roots of unity can be visualized as an exact rotations of a circle, justifying why they are called *cyclic groups*.

Of course, there is a shortcut to the above result. A corollary of the [Theorem 12](#) guarantees that for a rational number other than 0, 1, and -1 , its arctan is always an irrational multiple of π . In the above case, $\arctan(\frac{3}{4})$ should be an irrational multiple of π , which makes it never reach 1 no matter how many times it rotates.

Theorem 12 (Niven)

If $\theta \in [0, \frac{\pi}{2}]$ and both $\frac{\theta}{\pi}$ and $\sin \theta$ are rational, then $\sin \theta$ is either 0, 1, or $\frac{1}{2}$.

There is also a way more ‘algebraic’ proof as well.

Proof. Let $x = \frac{4}{5} + \frac{3}{5}i$ and claim it’s an n -th root of unity for some $n \in \mathbb{Z}^+$. Hence we get $(4 + 3i)^n = 5^n$. In the Gaussian integers, both sides factor into primes: since $(2 - i)^2 = 3 - 4i$, we have $4 + 3i = i(2 - i)^2$, while $5 = (2 + i)(2 - i)$. Raising each to the n -th power gives the following.

$$i^n(2 - i)^{2n} = (2 + i)^n(2 - i)^n$$

Note that $(2 + i)$ and $(2 - i)$ are non-associate primes. From here, the right hand side has $(2 + i)$ as a prime n times, while the left hand side does not contain it. Recall that a prime factorization by Gaussian integers must be unique up to units. As $n \neq 0$ by the [Definition 4](#), this is a contradiction. Therefore, x is not an n -th root of unity for any positive integer n . $\square$

From here, we have two different sets T and μ_∞ . However, as you might have already guessed, μ_∞ is a subset of T . If x is a solution to $x^n = 1$ for some positive integer n , we have $|x^n| = |1|$ by taking absolute values on each side, from which we get $|x^n| = |x|^n = 1$. The only real solution to it is $|x| = 1$.

In fact, μ_∞ forms a *subgroup* of T under multiplication. Moreover, every μ_n is itself a subgroup of μ_∞ and called *cyclic* as every element of μ_n can be written as an exponentiation of a *primitive n -th root of identity*. Finally, T , as you might have imagined, is called the *circle group*, and is also a subgroup of a much larger group $U(\mathbb{C})$, called the *unit group of the complex ring*.

$$\mu_n \leq \mu_\infty \leq T \leq U(\mathbb{C})$$

End.

You may now submit your work.

Optional evaluation sheet

Unlike PCSA, submitting the sheet is **optional** for homeworks.

For each question so far, you can evaluate it based on the following criteria:

- If you failed to understand the question, mark $\times$ in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark $?$ in the Evaluation column.
- If you understood the question and are sure you have solved it, mark $\circ$ in the Evaluation column.

No.	Evaluation	No.	Evaluation
Q1		Q7	
Q2		Q8	
Q3		Q9	
Q4		Q10	
Q5		Q11	
Q6			

There are these additional questions as well; feel free to answer them.

- (a) Which was the hardest for you, among those you have solved? _____
- (b) Which seemed to be the hardest for you, among those you haven't solved? _____
- (c) Which was your favorite question, if you had to choose one? _____